MPUMALANGA MINING SOLUTIONS (MMS)

TECHNOLOGY FRAMEWORK & ARCHITECTURE DOCUMENT

Version 1.0

PURPOSE

This document defines the approved technology stack, frameworks, libraries, development standards, architecture patterns, and implementation requirements for the MMS Digital Ecosystem.

The objective is to ensure consistency, scalability, maintainability, performance, security, and long-term growth.

Every AI agent, developer, designer, and contributor must follow this document.

DEVELOPMENT PHILOSOPHY

The MMS platform must be:

Fast

Scalable

Secure

Mobile First

SEO Optimized

AI Powered

Cloud Native

Component Driven

Accessible

Maintainable

Production Ready

FRONTEND FRAMEWORK

Primary Framework:

Next.js 16

Reason:

Server Components

Server Actions

SEO Friendly

Excellent Performance

Vercel Optimization

Modern React Architecture

Enterprise Scalability

UI FRAMEWORK

React 19

Reason:

Industry Standard

Large Ecosystem

Component Architecture

Future Proof

Excellent Developer Experience

LANGUAGE

TypeScript

Reason:

Type Safety

Maintainability

Enterprise Development Standards

Better AI Development Experience

Reduced Bugs

STYLING FRAMEWORK

Tailwind CSS

Reason:

Rapid Development

Consistency

Responsive Design

Modern Utility System

Excellent Next.js Integration

Maintainable Design System

COMPONENT LIBRARY

shadcn/ui

Reason:

Modern

Accessible

Customizable

Production Ready

Excellent Tailwind Integration

Enterprise Grade Components

ICON SYSTEM

Lucide React

Reason:

Modern

Consistent

Lightweight

Developer Friendly

Scalable

Professional Appearance

ANIMATION FRAMEWORK

Framer Motion

Reason:

Smooth Animations

Modern UX

Professional Feel

Page Transitions

Interactive Components

Premium User Experience

STATE MANAGEMENT

Primary:

React Server Components

Server Actions

Secondary:

Zustand

Use only when client-side state becomes necessary.

Avoid unnecessary complexity.

BACKEND FRAMEWORK

Next.js Server Actions

Reason:

Simplified Architecture

Reduced API Boilerplate

Better Security

Improved Performance

Modern Development Workflow

DATABASE

Supabase PostgreSQL

Reason:

Managed Infrastructure

Scalability

Security

Realtime Features

Enterprise Database

Reliable Backups

AUTHENTICATION

Supabase Auth

Features:

Email Authentication

Magic Links

Password Authentication

Session Management

Role Based Access Control

Future Social Authentication

STORAGE

Supabase Storage

Purpose:

Student Documents

Certificates

Course Materials

PDF Files

Application Forms

Images

Marketing Assets

Training Resources

SECURITY

Supabase Row Level Security

Mandatory

Every table must implement:

User Isolation

Role Permissions

Administrative Permissions

Data Protection

POPIA Compliance

EMAIL SYSTEM

Resend

Purpose:

Transactional Emails

Marketing Emails

Newsletters

Notifications

Automated Communication

Student Communication

Administrative Communication

EMAIL TEMPLATE SYSTEM

React Email

Purpose:

Professional Templates

Reusable Components

Brand Consistency

Responsive Email Design

AI SYSTEM

Primary Provider:

NVIDIA APIs

Purpose:

Virtual Receptionist

Student Support

Course Assistance

Content Assistance

Document Assistance

Lead Qualification

Administrative Automation

AI MODEL RECOMMENDATIONS

Receptionist Agent:

GPT-OSS-120B

Course Support Agent:

MiniMax M2.7

General Assistance:

DeepSeek

Future Expansion:

Agentic Multi-Agent Architecture

SEARCH SYSTEM

Supabase Full Text Search

Future:

Vector Search

Semantic Search

AI Knowledge Retrieval

DOCUMENT GENERATION

PDF Generation:

React PDF

Purpose:

Invoices

Receipts

Certificates

Enrollment Letters

Acceptance Letters

Reports

PAYMENT INTEGRATION

Phase 1

Manual EFT

Proof Of Payment Upload

Administrative Verification

Phase 2

PayFast

Paystack

Ozow

Yoco

ANALYTICS

Google Analytics

Purpose:

Traffic Tracking

Conversions

User Behavior

Campaign Performance

USER BEHAVIOR ANALYTICS

Microsoft Clarity

Purpose:

Heatmaps

Session Recordings

Behavior Analysis

UX Improvements

SEO FRAMEWORK

Next.js Metadata API

Structured Data

Schema Markup

Open Graph

Twitter Cards

XML Sitemap

Robots.txt

Local SEO

FORM MANAGEMENT

React Hook Form

Zod Validation

Reason:

Performance

Validation

Developer Experience

Type Safety

FILE UPLOADS

Supabase Storage

Validation Required

Accepted Types:

PDF

PNG

JPG

JPEG

Maximum Size Limits Required

Virus Protection Recommended

LEARNING MANAGEMENT SYSTEM

Architecture:

Custom LMS

Do not use Moodle.

Do not use WordPress LMS.

Do not use LearnDash.

Reason:

Full Control

Modern UX

Better Integration

Custom Features

LEARNING CONTENT TYPES

Video Lessons

Embedded YouTube Videos

PDF Notes

Presentations

Assignments

Assessments

Quizzes

Interactive Learning

Downloads

Progress Tracking

Certificates

ACCESS CONTROL SYSTEM

Student Access Depends On:

Enrollment Status

Payment Status

Administrative Approval

Course Expiration

Subscription Period

DASHBOARDS

Student Dashboard

Admin Dashboard

Facilitator Dashboard

Future Employer Dashboard

Each dashboard must have dedicated permissions.

NOTIFICATION SYSTEM

Email

In-App Notifications

Future:

WhatsApp Integration

SMS Integration

Push Notifications

API ARCHITECTURE

REST API

Server Actions

Future:

GraphQL

Agent APIs

Automation APIs

PROJECT STRUCTURE

/app

/components

/features

/lib

/actions

/hooks

/types

/utils

/emails

/public

/styles

Keep architecture modular.

Avoid monolithic development.

DEPLOYMENT

Platform:

Vercel

Environment Strategy:

Development

Staging

Production

Use environment variables.

Never expose secrets.

PERFORMANCE TARGETS

Page Load:

Under 2 Seconds

Lighthouse:

95+

Accessibility:

95+

SEO:

100

Best Practices:

100

Mobile Performance:

95+

ACCESSIBILITY

WCAG 2.1

Keyboard Navigation

Screen Reader Support

High Contrast Design

Accessible Forms

Accessible Components

DESIGN SYSTEM

Colors:

Gold #D9A400

Silver #C0C0C0

Black #111111

Surface #1A1A1A

White #F7F7F7

Typography:

Headings: Bebas Neue

Secondary: Montserrat

Body: Inter

DEVELOPMENT PRINCIPLE

Every feature must answer:

Does this improve student experience?

Does this improve trust?

Does this reduce administrative work?

Does this improve conversion?

Does this improve scalability?

If the answer is no, reconsider implementation.

FINAL OBJECTIVE

Build a modern digital-first mining training ecosystem capable of operating efficiently without a physical office while maintaining the appearance, professionalism, and operational standards of a large national training institution.

The platform must be capable of supporting:

Thousands of students

Multiple facilitators

Multiple courses

Online learning

Administrative automation

AI-powered assistance

Future accreditation requirements

Long-term business growth